

International Economic Relations

ECON 308

Labor Productivity and Comparative Advantage: The Ricardian Model
(Chapter 3 Part 1)

Instructor: Muhebullah Karimzada

Learning Objectives

3.1 Explain how the **Ricardian model**, the most basic model of international trade, works and how it illustrates the principle of **comparative advantage**.

3.2 Demonstrate **gains from trade** and refute common fallacies about international trade.

3.3 Describe the empirical evidence that wages reflect productivity and that trade patterns reflect relative productivity.

Introduction

- Differences across countries are a key reason why trade occurs:
 - The Ricardian model examines differences in the **productivity of labor** (due to differences in **technology**) between countries.
 - The Specific Factors model and the Heckscher-Ohlin model examine differences in **labor, labor skills, physical capital, land, or other factors of production** between countries.
- Trade may also arise due to economies of scale (larger scale of production is more efficient).

The Concept of Comparative Advantage (1 of 6)

- The opportunity cost of producing something measures the cost of not being able to produce something else with the resources used.
- Comparative advantage will be determined by comparing opportunity costs across countries.

The Concept of Comparative Advantage (2 of 6)

- A simple example with roses and computers explains the intuition behind the concepts of opportunity cost and comparative advantage in the Ricardian model.
- For example, suppose a limited number of workers could produce either roses or computers.
 - The opportunity cost of producing computers is the amount of roses not produced.
 - The opportunity cost of producing roses is the amount of computers not produced.

The Concept of Comparative Advantage (3 of 6)

- Suppose that in the United States, 10 million roses could be produced with the same resources as 100,000 computers.
- Suppose that in Colombia, 10 million roses could be produced with the same resources as 30,000 computers.
- Colombia has a lower opportunity cost of producing roses: has to start producing fewer computers in order to free up resources to make a rose.

The Concept of Comparative Advantage (4 of 6)

- A country has a **comparative advantage** in producing a good if the opportunity cost of producing that good is lower in the country than in other countries.
 - The United States has a comparative advantage in computer production.
 - Colombia has a comparative advantage in rose production.

The Concept of Comparative Advantage (5 of 6)

- Suppose initially that Colombia produces computers and the United States produces roses, and that both countries want to consume computers and roses.
- Can both countries be made better off?

Table 3.1 Hypothetical Changes in Production

	Million Roses	Thousand Computers
United States	−10	+100
Columbia	+10	−30
Total	0	+70

The Concept of Comparative Advantage (6 of 6)

- When countries specialize in production in which they have a comparative advantage, more goods and services can be produced and consumed.
 - Have the United States stop growing roses and use those resources to make 100,000 computers instead. Have Colombia stop making 30,000 computers and grow roses instead.
 - If produce goods in which have a comparative advantage (the United States produces computers and Colombia produces roses), they could still consume the same 10 million roses, but

could consume $100,000 - 30,000 = 70,000$ more computers.

A One-Factor Economy (1 of 4)

- We formalize these ideas by constructing a one-factor Ricardian model using the following assumptions:

A One-Factor Economy (2 of 4)

1. Labor is the only factor of production.
2. Labor productivity varies across countries due to differences in technology, but labor productivity in each country is constant.
3. The supply of labor in each country is constant.
4. Two goods: wine and cheese.
5. Competition allows workers to be paid a wage equal to the value of what they produce, and allows them to work in the industry that pays the higher wage.
6. Two countries: home and foreign.

A One-Factor Economy (3 of 4)

- A **unit labor requirement** indicates the constant number of hours of labor required to produce one unit of output.
 - a_{LC} is the unit labor requirement for cheese in the home country. a_{LC} hours of labor produce one pound of cheese in the home country.
 - a_{LW} is the unit labor requirement for wine in the home country. a_{LW} hours of labor produce one gallon of wine in the home country.
- A high unit labor requirement means low labor productivity.
 - Labor productivity is how much output one hour of labor creates.

A One-Factor Economy (4 of 4)

- Labor supply L indicates the total amount of labor resources—the number of hours worked (a constant parameter).
- a_{LC} indicates the amount of labor required for each pound of cheese produced (a constant).
- Cheese production Q_C indicates how many total pounds of cheese that the home country produces.
- a_{LW} indicates the amount of labor required for each gallon of wine produced (a constant).
- Wine production Q_W indicates how many total gallons of wine that the home country produces.

Production Possibilities (1 of 6)

- The **production possibility frontier** (PPF) of an economy shows the **maximum** amount of a goods that can be produced for a fixed amount of resources.
- The PPF of the home economy is

$$a_{LC}Q_c + a_{LW}Q_w \leq L$$

Production Possibilities (2 of 6)

- Maximum home cheese production is

$$Q_c = \frac{L}{a_{LC}} \text{ when } Q_w = 0.$$

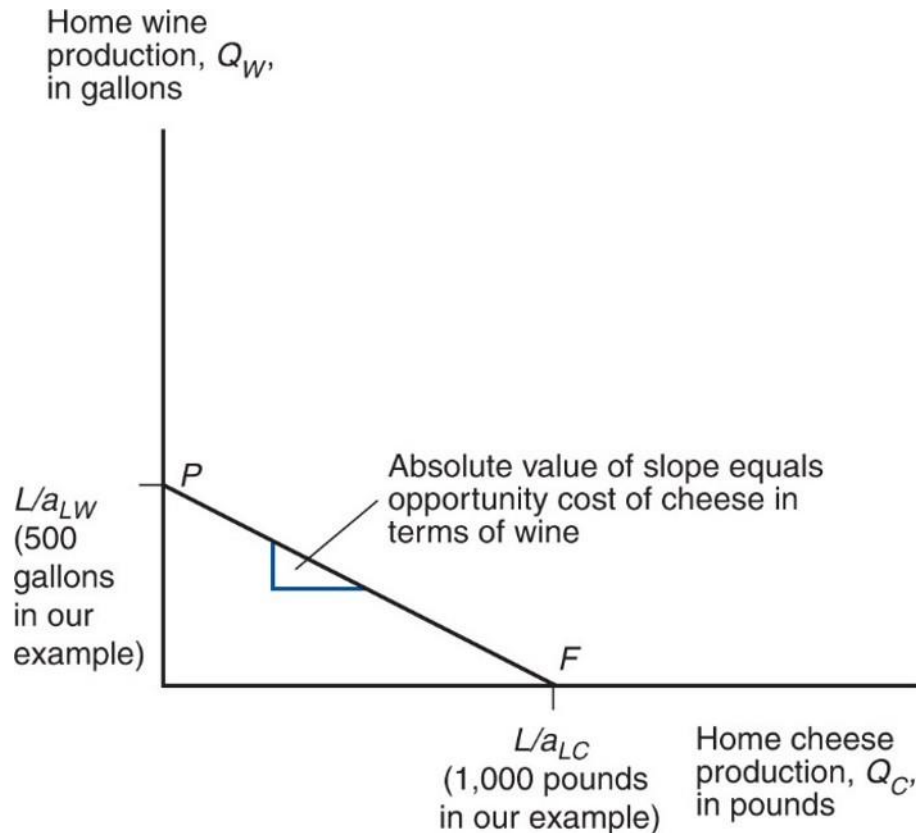
- Maximum home wine production is

$$Q_w = \frac{L}{a_{LW}} \text{ when } Q_c = 0.$$

Production Possibilities (3 of 6)

- For example, suppose that the home economy's labor supply is 1,000 hours.
 - $a_{LC} = 1 \text{ hour / lb}$, so one hour of labor produces one pound of cheese in the home country.
 - $a_{LW} = 2 \text{ hours / gallon}$, so two hours of labor produces one gallon of wine in the home country.
- The PPF equation $a_{LC}Q_C + a_{LW}Q_W \leq L$ becomes $Q_C + 2Q_W \leq 1,000$.
- Maximum cheese production is 1,000 pounds.
- Maximum wine production is 500 gallons.

Figure 3.1 Home's Production Possibility Frontier



The line PF shows the maximum amount of cheese Home can produce given any production of wine, and vice versa.

Production Possibilities (4 of 6)

- The opportunity cost of cheese is how many gallons of wine Home must stop producing in order to make one more pound of cheese:

$$\frac{a_{LC}}{a_{LW}}$$

- The opportunity cost is constant because the unit labor requirements are both constant.
- The opportunity cost of cheese appears as the absolute value of the slope of the PPF.

$$Q_W = \frac{L}{a_{LW}} - \left(\frac{a_{LC}}{a_{LW}} \right) Q_C$$

Production Possibilities (5 of 6)

- Producing an additional pound of cheese requires a_{LC} hours of labor.
- **Each** hour devoted to cheese production could have been used instead to produce an amount of wine equal to

$$1 \text{ hour} / (a_{LW} \text{ hours/gallon of wine})$$

$$= \left(\frac{1}{a_{LW}} \right) \text{gallons of wine}$$

Production Possibilities (6 of 6)

- For example, if one hour of labor is moved to cheese production, that additional hour could have produced

$$1 \text{ hour} / (2 \text{ hours/gallon of wine}) = \left(\frac{1}{2} \right) \text{ gallon of wine.}$$

- Opportunity cost of producing one pound of cheese is

$$\frac{1}{2} \text{ gallon of wine not produced.}$$

Relative Prices and Supply (1 of 7)

- P_C is the price of cheese; P_W is the price of wine.
- W_C is the wage paid to workers who make cheese, and W_W is the wage paid to workers who make wine.
- Due to competition in the labor and goods markets:
 - Hourly wages of cheese makers will equal the value of the cheese produced in an hour:

$$W_C = \frac{P_C}{a_{LC}}$$

- Hourly wages of wine makers will equal the value of the wine produced in an hour:

$$W_W = \frac{P_W}{a_{LW}}$$

Relative Prices and Supply (2 of 7)

- Workers will choose to work in the industry that pays the higher wage.
- If the price of cheese relative to the price of wine exceeds the opportunity cost of producing cheese

$$\frac{P_C}{P_W} > \frac{a_{LC}}{a_{LW}},$$

- Then the wage paid when making cheese will exceed the wage in wine

$$W_C = \frac{P_C}{a_{LC}} > \frac{P_W}{a_{LW}} = W_W.$$

- So workers will make only cheese (the economy specializes in cheese production).

Relative Prices and Supply (3 of 7)

- If the price of cheese relative to the price of wine is less than the opportunity cost of producing cheese

$$\frac{P_C}{P_W} < \frac{a_{LC}}{a_{LW}},$$

- Then the wage in cheese will be less than the wage in wine

$$W_C = \frac{P_C}{a_{LC}} < \frac{P_W}{a_{LW}} = W_W.$$

- So workers will make only wine (the economy specializes in wine production).

Relative Prices and Supply (4 of 7)

- If the price of cheese relative to the price of wine equals the opportunity cost of producing cheese

$$\frac{P_C}{P_W} = \frac{a_{LC}}{a_{LW}},$$

- Then the wage in cheese will equal the wage in wine

$$W_C = \frac{P_C}{a_{LC}} = \frac{P_W}{a_{LW}} = W_W.$$

- So workers will be willing to make both wine and cheese.

Relative Prices and Supply (5 of 7)

- For example, suppose cheese sells for $P_C = \$4 / \text{pound}$, and wine sells for $P_W = \$7 / \text{gallon}$.

- Wage paid producing cheese is

$$\frac{P_C}{a_{LC}} = (\$4 / \text{pound})(1 \text{ pound} / \text{hour}) = \$4 / \text{hour}.$$

- Wage paid producing wine is

$$\frac{P_W}{a_{LW}} = (\$7 / \text{gallon})(\frac{1}{2} \text{ gallon} / \text{hour}) = \$3.50 / \text{hour}.$$

- Workers would be willing to make only cheese (the relative price of cheese $4 / 7$ exceeds the opportunity cost of cheese $\frac{1}{2}$).

Relative Prices and Supply (6 of 7)

- If the price of cheese drops to $P_C = \$3 / \text{pound}$:

- Wage paid producing cheese drops to

$$\frac{P_C}{a_{LC}} = (\$3 / \text{pound})(1 \text{ pound} / \text{hour}) = \$3 / \text{hour}.$$

- Wage paid producing wine is still $\$3.50 / \text{hour}$ if price of wine is still $\$7 / \text{gallon}$.

- Now workers would be willing to make only wine (the relative

price of cheese $\frac{3}{7}$ is now less than the opportunity cost of cheese $\frac{1}{2}$).

Relative Prices and Supply (7 of 7)

- If the home country wants to consume both wine and cheese (in the absence of international trade), relative prices must adjust so that wages are equal in the wine and cheese industries.
- If $\frac{P_C}{a_{LC}} = \frac{P_W}{a_{LW}}$, workers will not care whether they work in the cheese industry or the wine industry, so that production of both goods can occur.
- Production (and consumption) of both goods occurs when the relative price of a good equals the opportunity cost of producing that good:

$$\frac{P_C}{P_W} = \frac{a_{LC}}{a_{LW}}$$